DATA 4401 Project 3

Fall 2026

Due: November 9/November 11

Note: the specific date for the project presentation may depend on whether we have project customers presenting on either of the dates.

Overview

You all (the entire class) are the data science division of a startup called **Winetitude** (yeah, I know), whose goal is to disrupt (of course) the wine review culture.

Wine industry interlude. The worldwide wine industry is a fairly unique business in terms of the product it produces and markets, and in terms of how its products find its way into the hands of paying consumers. Some of the unique factors about the wine industry are:

- **Vintages.** Most of the industry produces a new set of products (wines) every year. The wines are produced from the grapes harvested in a specific year (vintage), and thus the same brand of wine (e.g., Ancient Peaks Pearl Petit Sirah) is made out of different grapes, and often using somewhat different processes every year.
- **Variable quality across vintages.** Grape harvest quality/quantity is subject to change from year to year due to terroir and weather effects. As a result, two different vintages of the same product may be quite different.
- **Manufacturing.** Most wine producers are small/medium-sized wineries. Even when a winery is owned by a large conglomerate, wine branding reflects the individual winery identity and typically hides the conglomerate. As a result, there are a lot of wines on the market with names that are hard to remember even for dedicated wine connoisseurs, left alone for casual drinkers. Very few people have strong associations between a specific wine product (e.g., Dover Canyon Dusi Zinfandel) and its taste (and the difficulty of making such association is exasperated by the differences between the vintages).
- **Consumers.** Most people consume wine casually - with meals, or as part of entertainment/recreation. A small segment of the consumer market are connoisseurs and wine collectors. While their numbers are relatively small, they spend *a lot more money on wine* than casual consumers, and as a result, they

are the subject of targeted (marketing) efforts of the wine industry for the industry's higher priced/higher prestige wines.

- **Wine quality.** The diversity of wine, and the differences between multiple vintages of the same wine necessitate a more sophisticated effort for distributing information about the wines to consumers, first and foremost, to the dedicated consumers. Here, for the last 50 or so years, the wine industry coalesced on the following approach:
    - **wine scoring:** wines are scored on a 100 point system (in reality - on a point system with the operational range of 70-100), where higher scores are better, but the scale is rather non-linear. Scores of 90 and above indicate excellent wines, scores in low-to-mid 80s indicate low quality wines (scores below 80 are almost unheard of), leaving only a few values (87, 88, 89) for wines that one would consider to be "ok", "drinkable", "decent", and so on.
    - **wine scorers:** all scoring is done by professional wine tasters in controlled blind testing settings (the tasters/sommeliers usually know the types of wines being tasted, but not the specific brands/products). Wines are assessed on color, smell/aroma (otherwise referred to as "nose"), taste, mouthfeel, and aftertaste/lingering effects. A wine tasting results in two outcomes:
        - the **wine score** (see above) being assigned to the wine
        - the **wine review,** a short text review (see example below) written about the wine to accompany/justify the score.
    - **wine score distribution:** wine scores and reviews are published in a variety of magazines dedicated to wine, such as Wine Advocate, Wine Spectator, Wine Enthusiast (hope you are spotting a theme here), Decanter, Wine & Spirits, and so on. Each such magazine publishes, in every issue (usually at the back) a list of a few hundred wine reviews accompanied by the scores. Below is an example of one such review (from the dataset we will be using for this project):

"Grown on sandy soils, this cherry-hued Grenache offers up juicy watermelon and red fruit notes, with vanilla, white pepper and a dash of green herbs. The supple yet elegant palate boasts drying tannins balanced by lively acidity and a savory finish " (review for An Approach to Relaxation 2015 Sucette Grenache (Barossa) rated 89 by its reviewer).

- **Wine origin.** Wines are produced worldwide, and for the vast majority of wine products on the market, the key differentiator and the key piece of information that needs to be conveyed to the consumer is the wine provenance. This is done via a multi-layers ontology:

    - **Country.** At the highest level of the ontology is the country of origin. Very few wines use grapes from multiple countries, so almost every wine has a well-defined country of origin, such as France, Italy, Spain, Portugal, USA, Australia, Argentina, New Zealand, Chile, Germany, Romania, and others.
    - **Province.** In several countries, such as the US or Italy, the next step in the wine origin is the geographic subdivision (province, state, etc) of the country. For example, in the US, almost all wines are labeled with the state from which the grapes originated: California, Oregon, Washington, New York, etc.
    - **Appellation.** Perhaps the most important level of the wine origin ontology, this is the specific (and controlled by law) geographic designation outlining where the grapes for the wine were harvested. In France, this includes regions like Bordeaux, Burgundy, Rhone, Loire. In the US, in California we have Napa Valley, Sonoma Valley, Paso Robles, Monterrey County and so on.
    - **Smaller Viticultural Areas/subdomains.** Often, Appellations are subdivided into smaller regions to indicate even more precise origins of the grapes. Examples of this include Médoc, Sauternes and Saint-Emilion regions of Bordeaux in France, and Russian River Valley, Alexander Valley, Carneros subdivisions of Sonoma in California.
    - **Individual vineyards.** Some wines have grapes sourced completely from a single vineyard, and they may contain specific vineyard designation. For example "Dover Canyon Dusi Zinfandel" refers to a Zinfandel wine produced by the Dover Canyon winery with the grapes sourced from the Dusi vineyard in the Paso Robles region.
    - **Individual wineries.** Many if not all wines are branded by the winery that produced them. The winery name, unlike the vineyard name does not always signify the origin of the grapes, but rather identifies the wine maker/producer/manufacturer. Some wines are labeled as "Estate", which (informally) is used to specify that the grapes for the wine came from the vineyards owned by the winery (not all wineries own vineyards, not all vineyards are owned by wineries: viticulture - i.e., growing of the wine grapes, and winemaking are two separate types of business, although wineries often engage in both).
- **Wine varietals.** Finally, wines are distinguished by the grapes they are made of. These distinctions come at several levels:
    - **Wine color.** The most coarse distinction is between **red** and **white** (and now - **rosé**) wines. This, first and foremost indicates the actual color of the wine to the consumer, although, each "technical" wine color represents a range of actual colors (e.g., white wines can range in color from very

pale straw to deep amber). There are varieties of white and red grapes that are the primary source of wine color. Additionally, to produce red wines, the grape skins of red grape varietals are left to ferment with the juice, and they contribute to the coloring of the wine. Grape varietals like Chardonnay, Sauvignon Blanc, Viognier, Riesling are white; Zinfandel, Cabernet Sauvignon, Grenache, Syrah, Merlot are examples of red grapes.

- **Wine varietal.** Many wines are made out of one specific grape varietal, or out of one main grape varietal with small addition of one or more others. Typically, such wines are referred to by their sole or main grape varietal. For example if you see a Donati Family Winery Grenache, you are looking at a wine that is either made out of 100% Grenache grapes, or in which Grenache grapes dominate (usually compose 85% or more of the total grape input). Countries usually regulate when a wine can be called by a single varietal.
- **Blends.** Often, winemakers mix (blend) several grape varietals in significant proportions to obtain desired taste profile. Blends of 40% Syrah, 30% Grenache, and 30% Mourvedre (three grape varietals popular in the Rhone region of France) are very popular and are often referred to as "GSM blends". Some wines include blend designations (like "GSM" or "Cabernet Franc - Malbec") in their name. Some wines have a proper name "(Quinta dos Avidagos 2011 Avidagos Red) that does not specify the blend.
- **Other wine categories.** In addition to red/white/rosé table wine, there are several other wine categories that you may encounter in your work. Of these, the big ones are dessert and sparkling wines.
  - **Dessert wines** are produced either from late harvest grapes (these grapes start turning into raisins, retaining their sugars but losing water, thus making them much sweeter), or are made by adding sugar and neutral spirits to the fermented grape juice (these are known as ***fortified dessert wines*)**. These wines (such as Portuguese Porto wines, Spanish Jerez, or French Sauternes, or Canadian and German Ice Wines) have very sweet taste, and are typically not compared to table wines.
  - **Sparkling wines** like the ubiquitous Champagne (produced in the Champagne region of France), Cava (Spain) or Proseco (Italy) are made using a variation of what is known as *methode champenoise* - a way of fermenting the grape juice that introduces some level of carbonation into the bottle.

Winetitude and its goals. Winetitude is a startup whose goal is to democratize the cottage industry of wine reviews as it exists right now. Winetitude founders firmly believe that the current structure of wine scores and reviews - essentially relying on the taste experience of one person for a score that stays with the wine forever, and that may make or break a particular wine, is **severely flawed**. Winetitude founders believe that this approach is both prone to error/misjudgement of the merits of individual wines, and also opens the industry up to a level of corruption that can result in deliberate manipulation of wine scores to the benefit of one wine producer or another.

In their quest to establish a more transparent and more robust way of providing wine quality assessments, Winetitude is interested in analyzing the reviews produced by the current wine review cottage industry.

This is where you come in.

Dataset

For this project, you will be working with the **Wine Reviews** dataset available on Kaggle at this URL:

https://www.kaggle.com/datasets/zynicide/wine-reviews

The dataset contains upwards of 130,000 reviews presented three different ways (two CSV files, one of which appears to have around 150,000 records, but may contain a lot of repetitions, and one JSON file)[1]. Together with the reviews (typically a short text of about 30 to 60 words), the dataset provides information about the wine tasted, including its name, its variety/designation, and its geographic origin, the points it received, the identity of the author of the review (the person who tasted the wine). Understanding the full scope of the dataset is left to each team - the data is mostly self-explanatory, but the specific fields of the dataset describing the wine origin and designation need to be examine manually and through basic EDA which all of you are able to do by now.

Feature engineering and other data. The tasks you are asked to do primarily concern the analysis of the review text, which you will be vectorizing in a variety of ways according to the needs of your task. However, you are allowed to include different third-party datasets in your

[1] For the record, when setting up this project, and also when working with this dataset in DATA 3301, I used the JSON file for the data ingestion. I have not compared the two CSV files to each other to discover the differences - this - if you want to do it - is left up to you.

analysis if you feel that they may contribute to your ability to build accurate predictive or (in some cases) interpretable models. For example, it is well known, the weather affects wine quality. Following the footsteps of Orly Ashenfelter (some of you may have heard this name in DATA 301), you could decide to incorporate some weather/climate characteristics into your dataset.

Additionally, you can, and in some cases, will, engineer your own features from the data made available to you. For example, the vintage of the wine is not available in a separate attribute in this dataset, but can be found as part of each wine name. You should parse wine names to extract the vintage. Other information about the wine origin can also be aggregated, compartmentalized, engineered into additional attributes.

Finally, your **main feature engineering effort** will concentrate on the reviews. As mentioned, each review is a short text (1-3 sentences, rarely more than 60-65 words). Your goal is to turn the text into a set of features by vectorizing it using whatever techniques and feature extraction mechanisms you want. The end result of this process shall be a 2D matrix of feature weights for each review (e.g. a TF-IDF matrix) that you will primarily be using in your analysis. Some of your analysis may combine the feature weight matrix with additional attributes engineered from other sources, but as mentioned above, strictly speaking, for this assignment, ***this is NOT required.***

Tasks

Unlike other projects, where all teams in the class work on the same task, for this project, each team is going to work on an assignment of their own. Essentially, your entire class represents the data science division of Winetitude (they had enough investor capital to hire 35 data scientists in order to speedrun their analytical needs). This division is broken into **eight teams** (named after famous viticultural regions of the world) and each team has a specific task and target variable to predict/estimate/model.

Overall, every task comes with a similar structure. Each team is given one target variable (or one category of target variables) to work on modeling/predicting. Each team is also given a specific ontology of the dataset (sometimes in more detail, sometimes in less) that first identifies a small subset of reviews that need that need to be modeled, and provide the pathway for extending your modeling efforts to a larger and larger sets of reviews until you start building models for the entire dataset.

The targets and target assignments to teams are described below together with the specific ontologies to work through. Individual team rosters are announced in class.

Task Assignments

The outline of the task assignments is as follows:

1. **Team Bordeaux:** wine color
2. **Team Rhone:** wine review authorship
3. **Team Douro:** wine scores
4. **Team Veneto:** wine prices
5. **Team Rioja:** wine origins
6. **Team Napa:** wine vintages
7. **Team Sonoma:** wine varietals
8. **Team Paso Robles:** progress through years

Below, we discuss the eight team challenges, after which we provide additional guidelines on how to approach the problems.

Team Bordeaux: Predicting Wine Color from Reviews

Wine critics use different language when describing white and red wines. Some wine aromas and flavors are very likely to be found predominantly in red wine descriptions, while others - in white wine descriptions. Winetitude is interested in understanding the differences and the similarities between the language used to describe red and white wines.

To this extent:

- your team needs to establish which wines in the dataset are white and which are red. You DO NOT need to designate **every wine** as either white, red (or rosé in some cases, or other - if the wine is a dessert or a sparking one), but you do need to emerge is a sizeable and representative subset of wines where you are confident you know the wine color. Most of this can be derived from varietals (Cabernet Sauvignon is red, Chardonnay is white, etc.), and direct designations (such as Portuguese red), but for some wines you may need to be more subtle.
- your team needs to construct a series of classifiers that attempt to distinguish between red and white wine reviews (and, if you have enough samples, between red, white, rosé, sparkling, and dessert wines - the latter two are not wine colors, but are major wine categories where the language of reviews may also be different).

- determine both the accuracy/performance of your predictive models, as well as analyze and present to Winetitude the details on how red and white (you can concentrate here on these two categories) reviews are different, and also what common features they share.

Team Rhone: Predicting Wine Review Authorship

One of the key positions taken by Winetitude is that the wine scoring/review business is very biased (which may lead to errors of judgement or deliberate deception). To provide some credence to this position, your team is asked to determine/detect how hard/easy it is to discover the author of a review based on the text of the review. Please be aware that reviews may also be affected by compounding variables/latent factors - e.g., reviewer assignments are not independent of geography and time (the same person may be reviewing all or majority of wines from a specific region in a specific year). These are important considerations as certain words (e.g., geographic locations) appearing in reviews are more of an indicator of wine origin rather than individual styles. Still, your task is to see if you can model individual reviewer writing style.

To this extent:

- select a group of reviewers you will be modeling. There is a heavy imbalance in the dataset - some reviewers wrote thousands of reviews, some - very few. Set up a review count threshold, and select a non-trivial set of reviewers whose work you will be analyzing.
- for the reviewers you selected, train an authorship attribution classifier. You have the choice of whether you want to build multiple binary "one-vs-the-world" classifiers for separating reviews of one person from all others, or a single multilabel "all-vs-all" classifier.
- analyze and present to Winetitude the writing styles of select reviewers. Determine which reviewers have more distinct (easy-to-detect) style, and which have more generic, harder-to-detect style.

Team Douro: Predicting Wine Scores

Wine scores are the key outcome of the wine tasting/reviewing process. As we discussed in the overview of the wine industry, wine scores have lasting effects on sales and reputation of wines and their producers. Winetitude is very interested in being able to detect wine scores from the text of the reviews. It is well-known that certain

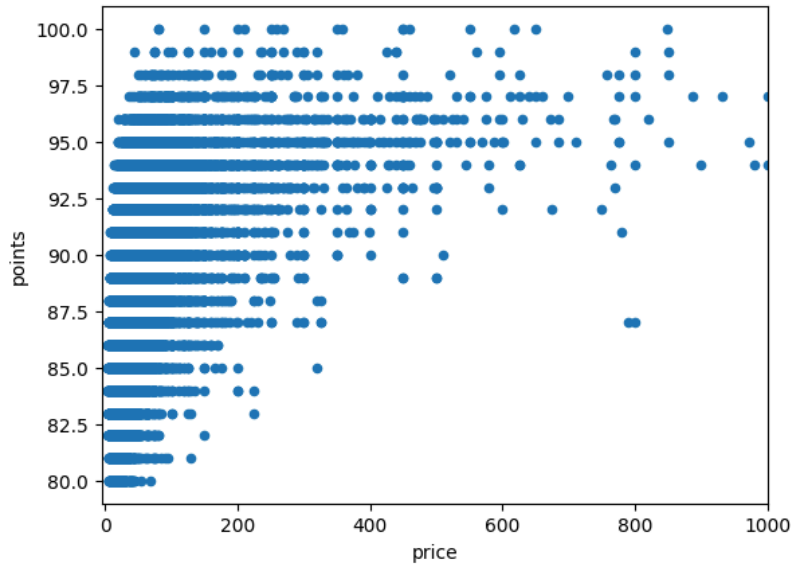
words in the reviews code for positive impressions of the wine, while other terms address explicitly, or imply defects.

To this extent:

- develop a simple classifier that determines if a review is positive, or negative, and a somewhat more involved three-level classifier that adds neutral reviews into the mix. Use information provided about the wine scores (you are also welcome to study wine literature for this) to determine your thresholds for positive, neutral, and negative reviews. For the binary classifier, neutral reviews can be excluded from consideration.
- develop and evaluate a regression model that predicts the specific wine score based on the review text. You can leverage the insight you got by building the classifier.
- determine and present to Winetitude the factors/language features that are biggest tells of positive or negative reviews.

Team Veneto: Predicting Wine Prices

A picture is worth a thousand words, and below is a (somewhat truncated by price for the sake of clarity) scatterplot of price vs. the score assigned to the wine:



Simply put, suspicions that perception of price of the wine being tasted (and this is one thing reviewers may often have some idea about, based on the context of their tasting experience) affects the review and the score have been expressed on multiple

occasions.[2] This issue did not escape Wintitude's attention, and the company is asking your team to consider whether the text of review is predictive of the price of wine.

To this extent:

- build models that predict the price of wine based on the text of the review. Your team can entertain both regression models to predict the exact price of wine, as well as classifiers to predict one of several price ranges into which a wine falls. Price range determination is left up to your team, although I recommend searching for literature that segments the wine market by price (this is a common thing - upselling the buyers into the next price category for wine is one of the biggest industry quests, and thus wine industry has a really good idea of what these price brackets are).
- Determine what language features are most reflective of the price/price category of the wine and present this insight to Winetitude.

Team Rioja: Predicting Wine Origin

As discussed in the introduction, wine origin is a highly regulated, and a highly influential notion. It is thus important to Winetitude to understand whether wine reviews for wines of different origin differ in the language. Your team is tasked with establishing to what degree this takes place. One of the challenges your team has to overcome (similarly to Team Rhone) is the fact that review authorship and origin of wine are not independent in our dataset. Certain authors/tasters have responsibility for tasting/scoring wines of specific origins (just check the distribution of authorship of Portuguese wines for example). Your team needs to build models that attempt to separate the difference in language used to describe wines of different origins from the differences that are due ot personal writing styles. Some ways of doing so may involve looking at the reviews of specific authors for wines from different countries/appellations.

In addition, certain domains of origin -- especially in Europe -- correlate heavily with grape varietals (e.g., Merlot is a Bordeaux grape, while Grenache is a Rhone grape) - you should be cognizant of this. In the US and Canada, this relationship is not as pronounced (Napa, Sonoma, Monterey, Santa Barbara, Paso Robles all feature a variety of wines from different domains of origin in France, Spain, Italy, Portugal and Germany).

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[2] I have some personal experience with this - I once participated in an analysis of wine scores, an analysis that identified that the price of wine (known to those whose scores we analyzed) accounted for around 30% of variance -- the highest factor available for us to consider (this work was actually performed by a student from an earlier DS Capstone, although outside of the Capstone context).

To this extent:

- develop classifiers that predict based on the text of the review the wine's country of origin. You may limit yourselves to one-vs-the world classifiers that predict a specific country, or explore multi-label classifiers. You can limit your classifiers to consider only countries with sufficient number of reviews in the dataset (there is a long tail of wines from lesser-known wine-producing countries which you can omit/ignore)
- develop classifiers that predict certain well-known domains of origin/appellations (such as Bordeaux, Burgundy, Rhone, Douro, Sonoma, Napa, etc). You can study which appellations are important, you can also use the information from the dataset to see which regions/appellations are prevalent in the dataset, and concentrate on those.
- determine what language features are associated with specific countries and domains of origin and present this information to Winetitude.

Team Napa: Predicting Wine Vintage from Reviews

We discussed that the wine vintage (year of the harvest) has great influence on the quality of wine - often, not just on the quality of individual wine, but one overall quality of wines from a specific region. This relationship may be more subtle than some of the other relationships that other teams are investigating, but it does not make it any less important. Your team's goal is to explore models that predict wine vintage based on the text of the review.

To this extent:

- add wine vintage as a feature into your dataset. This information needs to be instructed from the name of the wine (*title* attribute).
- explore a variety of models for telling different vintages apart, predicting the wine vintage based on the review. Classifiers are more appropriate than regression models, as vintage years are buckets, and the original relationship between vintage years is not really of as much interest as just plain ability to separate different vintages. Because quality of wine (and thus the overall gist of reviews) interacts with specific vintage year at a regional level (effect of weather/climate on wine in California and in New Zealand may be different for the same year), you should start with exploration of single-appelation/single geographic area models. Once you have robust models for smaller geographic areas, you can attempt generalizing them to a model for larger areas. Concentrate on areas and vintages where you have abundance of data to train the models.

- explore - if you want - whether adding data from external sources (e.g., weather data) improves your predictive power (note that weather data by itself may be considered a great discriminator of vintage years, so be careful with how you handle external sources. Your goal remains establishing a relationship between review text and vintage). One idea to explore is to use weather attributes as proxies for vintage and try to predict them (you should not build regression models for vintage year, but you MAY explore regression models for average summer temperature, for example.)
- determine the strongest relationships between language features and vintages for select regions (and for any generalized models you were able to make sufficiently accurate) and present them to Winetitude.

Team Sonoma: Predicting Wine Varietal from Reviews

Your team's task is similar in nature to the task of Team Bordeaux, but Winetitude is asking you to produce insight for a more granular distinction between wine types. Essentially, review language differs not only between red and white wines in general, but -- because wines from different grapes have often drastically different taste profiles -- for wines of the same color but made from grapes of different varieties. Your team's goal is to establish what makes Chardonnay reviews different from Sauvignon Blanc reviews, or Merlot reviews different from Petit Sirah reviews.

To this extent:

- select several red and white varietals that are popular in the dataset. In addition to specific single-grape varietals, you can use designations such as "Portuguese Red" as a varietal class for the purposes of building your models.
- Train multiple classifiers to predict the specific grape varietal given a review. You can start by limiting your classifiers to consider wines from a single appellation (e.g., compare Bordeaux Merlots to Bordeaux Cabernet Sauvignons), and to consider binary separation (Merlot vs Cabernet Sauvignon, like in the example above). You can also explore multi-class classifiers (predict ANY Bordeaux varietal). After this you can switch to inter-regional differences (Rhone Syrah vs Burgundy Pinot Noir), and then generalize to the global market (all Cabernet Sauvignons vs all Merlots vs all Malbecs vs all Syrahs).
- determine what language features are associated with specific grape varietals (at a regional, or a more global level) and present this insight to Winetitude.

Team Paso Robles: Progress through the Years

Your team has a special assignment. Rather than concentrating on a specific predetermined target variable, you will be able to select one or more target variables for yourselves. They can be (and most likely - will be) repeats of the target variables pursued by other teams. What is different about your team is how Wintetitude wants you to perform the analysis.

As we discussed, the wine industry operates on an annual cycle. Every year, a new vintage is released (technically, releases may be from different vintages, e.g., a winery may release a 2024 white wine and a 2022 red wine at the same time), and older wines start disappearing from the shelves. Moving forward from the present, updates to the review dataset will come in batches. Winetitude wants you to build models (for target variables of your choice) via a process that takes into account the timing of data acquisition.

To this extent:

- select one or more target variables of interest to your team. Try selecting something straightforward first (like grape color or score).
- develop a model training process for the predictive models for your target variables that incorporates data in chronological order, and track the model's performance (and evolution) through the iteration steps. You may start with tracking a single region over time, and then extending your exercise to a larger set of data until you start building models to work with all the data made available. As mentioned above, information about wines from a specific vintage won't necessarily appear all at once (I really wish the dataset contained review timestamps, but it does not and figuring those out is out of scope for this project), but vintages are good proxies for when wines are released, so making the next vintage your new batch approximates what future data dumps will look like reasonably well.
- review the evolution of the reviews over time and determine if there are any significant, noticeable changes in the language that can be noted. Share this insight with Winetitude.

Data Modeling and Model Training

This section describes information about the project that is relevant for all teams.

Predictive and Interpretable modeling. First, each team received an assignment that contains both a predictive modeling, and an interpretable modeling component. For most of the project, the goal of each team is to develop the best predictive models. When this goal is accomplished, you can decide how to address the need to provide insight from interpretable modeling. You can choose to repeat the modeling with interpretable models, or to extract interpretation - where you can (think about the types of ML models you can use) from your best predictive models.

Choice of ML methods. The choice of ML methods for this project is left up to you. You can use whatever techniques you want. The only limitation that we set forth is:

- do not use LLMs-as-a-blackbox (or LLMs at all) as your model.

Your goal is to use a method, or multiple methods where you can exercise full fine-tuned control over the entire training process.

Model evaluation. Your team is responsible for picking the right model evaluation procedure (train-test-validation split protocol) and optimizing and satisficing metrics. Your team may also consider if the notion of fairness applies to any of your modeling efforts (note: we are not talking here about legally protected classes, since our units of observation are wine reviews, but rather we are asking whether something like Equal Odds criterion may be useful in assessing how your model performs on reviews with different metadata).

Bias-Variance Tradeoff and model training. The key part of this project is to enable each team to use Bias-Variance tradeoff as the driving factor behind model training process.

- Start training **all your models** on very small training sets. The idea behind it is that Winetitude wants you, at first, to limit your predictions to a very small subset of data selected based on some criteria (e.g., predict review polarity [positive, negative] for 2012 Santa Rita Hills Chardonnays), before moving to extend it to a larger set of observations (all Santa Rita Chardonnays, then all California Chardonnays, then all Charonnays, possibly with additional stops along the way).
- Explore fully the Bias-Variance tradeoff:
    - start with very simple models
    - modify the model and add attributes for as long as the model underfits
    - add data to training set when the model starts overfitting

    - rinse, repeat
    - consider when overfit needs to be addressed with regularization or dataset simplification

  Document your steps.

- Note when your models stop generalizing. E.g., you may discover that your predictive models can be quite accurate for a subset of data representing all California wine reviews, but the same model does not generalize well when you apply it to wines from Europe.

Communication. This is a unique situation, when communication between teams is not only allowed, but also **encouraged.** By this time, you've been on earlier project teams with people from several other Project 3 teams, so everyone should have some "social network" of connections. You all are working on different tasks, but you all are working towards the same overarching goal. As a result, if your team discovers something that may be of interest to another team (e.g., you investigated differences between different regions in France and discovered that this can be explained by wine varietals, and would like to consult with the team that studies wine varietals) -- **feel free to share the information and consult.** If any such information sharing leads to an important breakthrough for your team - your team shall acknowledge this in your final report's Acknowledgements section (see below).

This collaboration may involve some limited code sharing (after all, we are emulating data science division of a company that has a shared codebase). Specifically, if your team discovers the need to do data processing in a specific way that has already been earlier done by another team, I have no problem with you sharing the code. The only thing that should be **fully authored by each team** is all the model training/evaluation code.

Deliverables

Project Progress Review

Due: Monday, November 2

For this project, our project progress review shall consist of two parts. As the entire class is playacting the data science division of a company, we want to facilitate communication between different teams. As a result:

- First, each team will give an informal 5-minute presentation to the rest of the class. The presentation shall be structured around **one interesting thing your team learned** while working on the project. The presentation should be accompanied by a small (up to 10, but possibly significantly fewer) slide deck, and *can be given by a designated subset of the team* on behalf of the entire team. The target audience for these presentations are your fellow data scientists who are well aware of the data you are working with and of the overall nature of the project, but may have only a surface idea of the specific challenge your team is facing.
- After this, I will hold a short meeting with each team (this might stretch into November 4 if needed). In this meeting I'll ask additional questions about your team's work and overall project status, and will answer any questions from the team.

Presentation

Due: Monday, November 9

Unless we have Capstone customers coming that day, we plan to have project presentations on Monday, November 9. Each team gets 10 minutes to present its main results. In preparing the presentation, please consider the following:

- The audience for your presentation are Winetitude founders/technical leadership, as well as the rest of the company's data science division. The audience **is friendly.**
- While your presentation shall be self-contained (i.e., it should start with the specification of the exact challenge your team is addressing, since it is different from everyone else's), you can assume general knowledge of the dataset, and of the overall goals of the company. On the technical level, target your presentation at your fellow data scientists. The technical leadership of Winetitude will be able to understand.
- At the same time, make sure that the company founders, who are not computer or data scientists walk away from your presentation with a takehome message - so make sure your conclusions and key insight are presented in a way that is accessible by subject-matter experts (they understand the wine industry and wine review industry very well).
- The key goal of your presentation is to unveil unique insights your team discovered. Make it fun and interesting.

Written Report

Due: Tuesday, November 10

Turn in a project report that discusses:

- your team's problem and the overall approach your team took to addressing it
- any data cleaning/feature engineering/data preparation your team conducted
- your modeling approach - what ML methods you used, how you tuned your models, what metrics and evaluation procedures you selected.
- Your model training process - subsets of the data for which you built individual models, bias-variance tradeoff-based decisions, and final models you coalesced to.
- evaluation of your final models (accuracy results, etc…)
- any insight you gained about the specific language features that had strong associations with your target variables (and any other model interpretation work you conducted).
- acknowledgements - if you communicated with other teams and want to acknowledge their influence on your work. Also acknowledge any LLM/GenAI use in the project/report.

In addition, please include, as an appendix, the following:

- a short personal reflection of each team member on their contributions to the team and their personal take on the insight from your team's results
- an overall team assessment of how using bias-variance tradeoff for model development worked for your team. Where did it help? Where was it not useful?

Code

Due: Tuesday, November 10

Submit all the code you used (Python notebooks or files) on this project.

Peer Review Survey

A peer review survey will be shared with you after all deliverables are due and will be due a few days after. While the peer review survey responses are not scored and are not part of the project grade, they are ***a required component*** of the project (your grade will not be awarded w/o a submitted peer review survey).